

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2021-2022
FULL MARKS: 150

Chem 4241: Chemistry

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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- ✓ 1. a) What are the fundamental particles of an atom? Briefly describe them. 6
(CO2)
(PO1)
- b) Discuss Bohr's theory of hydrogen atom. What modifications were proposed by Sommerfeld and why? 9
(CO2)
(PO1)
- c) Derive the equation for calculating the energy of electron in the orbit of hydrogen atom and calculate energy of the electron in the 4th orbit. 10
(CO2)
(PO1)
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- ✓ 2. a) What do you understand by the dual nature of electron? Derive de Broglie's equation and show its implications. 6
(CO2)
(PO1)
- b) Derive Schrodinger's wave equation and show the significance of Ψ and Ψ^2 . 10
(CO2)
(PO1)
- c) What are quantum numbers? What do they signify? Describe the relationship between them. 9
(CO2)
(PO1)
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- ✓ 3. a) Write Henry's law and show the effect of temperature and pressure on the dissolution of gases in liquid. 8
(CO1)
(PO1)
- b) Show through mathematical derivation that the solubility of solids in liquids is generally endothermic in nature and that the curve of solubility against temperature is exponential. 8
(CO1)
(PO1)
- c) Define osmosis, osmotic pressure, and reverse osmosis. Mention their uses. Deduce an equation to establish the relationship between molecular weight of a solute and osmotic pressure. 9
(CO1)
(PO1)
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- ✓ 4. a) Define vapor pressure, boiling point, and colligative properties. Name the colligative properties. 6
(CO1)
(PO1)
- b) Derive an equation showing the relationship between lowering of vapour pressure and molecular weight of an unknown solute in an ideal solution. 12
(CO1)
(PO1)

c) The vapour pressure of ether [MW = 74] at 250°C is 440mm of Hg. When 6.5g of an unknown solute is dissolved in 50g ether, the vapour pressure of the solution is 410mm of Hg. What is the Molecular Weight (MW) of the unknown solute?

5. a) Define and classify solution. Name the units of concentration and describe molarity and normality with examples.

b) What is Critical Solution Temperature (CST)? Draw and explain the CST diagram for the phenol-water system. What is the application of this diagram?

c) 20gm of NaCl is dissolved in 100ml water. Find out the molarity (M) and the molality (m) of the solution. The density of the solution is 1.06gm/cc (MW of NaCl = 58.5).

6. a) State and explain Lowry-Bronsted and Arrhenius concepts of acids and bases. Mention the salient features of the concepts with suitable examples.

b) What is a buffer solution? Briefly describe the types and mode of operation of buffer solutions.

c) Discuss the meaning of the terms pH and K_w and show how they are interrelated.

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